LETTER

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Highly efficient organic light-emitting diodes from delayed fluorescence

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The inherent flexibility afforded by molecular design has accelerated the development of a wide variety of organic semiconductors over the past two decades. In particular, great advances have been made in the development of materials for organic light-emitting diodes (OLEDs), from early devices based on fluorescent molecules[1] to those using phosphorescent molecules[2,3]. In OLEDs, electrically injected charge carriers recombine to form singlet and triplet excitons in a 1:3 ratio[1]; the use of phosphorescent metal–organic complexes exploits the normally non-radiative triplet excitons and so enhances the overall electroluminescence efficiency[2,3]. Here we report a class of metal-free organic electroluminescent molecules in which the energy gap between the singlet and triplet excited states is minimized by design[4], thereby promoting highly efficient spin up-conversion from non-radiative triplet states to radiative singlet states while maintaining high radiative decay rates, of more than 10^6 decays per second. In other words, these molecules harness both singlet and triplet excitons for light emission through fluorescence decay channels, leading to an intrinsic fluorescence efficiency in excess of 90 per cent and a very high external electroluminescence efficiency, of more than 19 per cent, which is comparable to that achieved in high-efficiency phosphorescence-based OLEDs[3].

The recombination of holes and electrons can produce light, in a process referred to as electroluminescence. Electroluminescence in organic materials was first discovered in 1953 using a cellulose film doped with acridine orange[5], and was developed in 1963 using an anthracene single crystal connected to high-field carrier injection electrodes[1]. Electrical charge carriers of both polarities were injected into the organic layers, and the subsequent carrier transport and recombination produced blue electroluminescence originating from singlet excitons; that is, fluorescence. According to spin statistics, carrier recombination is expected to produce singlet and triplet excitons in a 1:3 ratio[6,7], and this ratio has been examined for many molecular systems[8–12]. The singlet excitons produced decay rapidly, yielding prompt electroluminescence (fluorescence). Two triplet excitons can combine to form a singlet exciton through triplet–triplet annihilation, which results in delayed electroluminescence (delayed fluorescence). Direct radiative decay of triplet excitons results in phosphorescence, but usually occurs only at very low temperatures in conventional organic aromatic compounds. The first demonstration of phosphorescent electroluminescence using ketocoumarin derivatives in 1990[13]. However, the very faint electroluminescence was observed only at 77 K, and with difficulty, and was assumed to be virtually useless even if included in rare-earth complexes, which should also involve both singlet and triplet excitons in electrical excitation[14]. In 1999, efficient electrophosphorescence was first demonstrated using iridium phenylpyridine complexes that achieve an efficient radiative decay rate of $\sim 10^6\ s^{-1}$ by taking advantage of the strong spin–orbit coupling of iridium[2]. An internal electroluminescence efficiency of almost 100% was achieved[3], providing convincing evidence that OLED technology can be useful for display and lighting applications.

In the work reported here, we used a novel pathway to attain the greatest possible electroluminescence efficiency from simple aromatic compounds that exhibit efficient thermally activated delayed fluorescence (TADF) with high photoluminescence efficiency. Figure 1a shows the energy diagram of a conventional organic molecule, depicting singlet (S_1) and triplet (T_1) excited states and a ground state (S_0). It was previously assumed that the S_1 level was considerably higher in energy than the T_1 level, by 0.5–1.0 eV, because of the electron exchange energy between these levels. However, we found that careful design of organic molecules can lead to a small energy gap (ΔE_{ST}) between S_1 and T_1 levels[4,15]. Correspondingly, a molecule with efficient TADF requires a very small ΔE_{ST} between its S_1 and T_1 excited states, which enhances $T_1 \rightarrow S_1$ reverse intersystem crossing (ISC). Such excited states are attainable by intramolecular charge transfer within systems containing spatially separated donor and acceptor moieties[4]. The critical point of this molecular design is the combination of a small ΔE_{ST}, of $\lesssim 100$ meV, with a reasonable radiative decay rate, of $>10^6\ s^{-1}$, to overcome competitive non-radiative decay pathways, leading to highly luminescent TADF materials. Because these two properties conflict with each other, the overlap of the highest occupied molecular orbital and the lowest unoccupied molecular orbital needs to be carefully

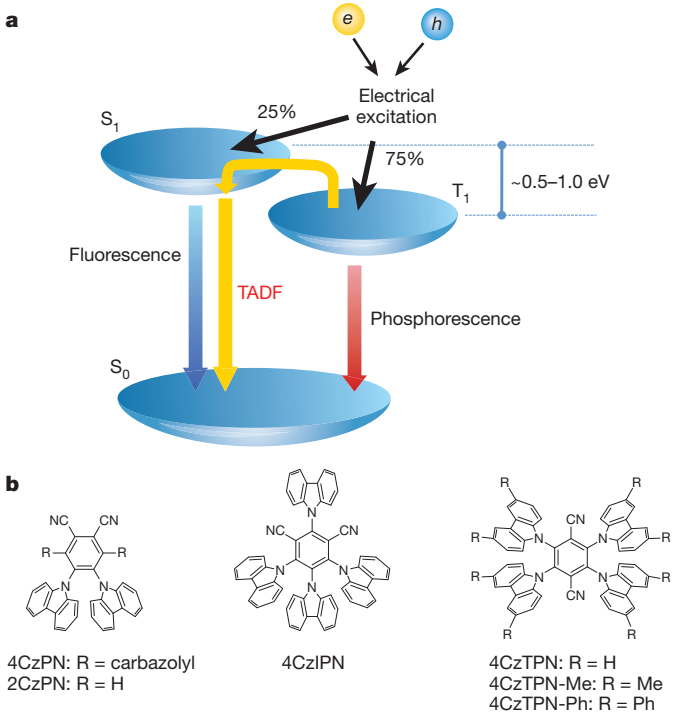


Figure 1 | Energy diagram and molecular structures of CDCBs. **a**, Energy diagram of a conventional organic molecule. **b**, Molecular structures of CDCBs. Me, methyl; Ph, phenyl.

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balanced. Furthermore, to enhance the photoluminescence efficiency of a TADF material, the geometrical change in molecular conformation between its S_0 and S_1 states should be restrained to suppress non-radiative decay. Limited orbital overlap generally results in virtually no emission, as has been shown in benzophenone derivatives. Therefore, it was previously assumed that a high photoluminescence efficiency could never be obtained from molecules with a small ΔE_{ST}. Here we demonstrate that it is possible to realize a high photoluminescence efficiency and a small ΔE_{ST} simultaneously.

We designed a series of highly efficient TADF emitters based on carbazolyl dicyanobenzene (CDCB), with carbazole as a donor and dicyanobenzene as an electron acceptor (Fig. 1b). Because the carbazolyl unit is markedly distorted from the dicyanobenzene plane by steric hindrance, the highest occupied molecular orbital and the lowest unoccupied molecular orbital of these emitters are localized on the donor and acceptor moieties, respectively, leading to a small ΔE_{ST}. Moreover, the dicyanobenzene and carbazolyl groups are important in obtaining a high photoluminescence efficiency and various emission colours, respectively. Dicyanobenzene derivatives are known to alter their chemical bonds in excited states, changing their electronic properties[16]. Density functional theory (DFT) calculations predicted that using CDCBs would have the following advantages. The cyano groups suppress both non-radiative deactivation and changes in the geometries of the S_1 and T_1 states of CDCBs, leading to a high quantum efficiency. Conversely, the emission wavelengths of CDCBs should be easy to tune by changing the electron-donating ability of the peripheral groups, which can be altered by changing the number of carbazolyl groups or introduced substituents. Such molecular design should allow us to achieve not only highly efficient TADF but also a wide range of emission colours.

We synthesized CDCBs from commercially available starting materials in a one-step reaction. Palladium or other rare-earth-metal catalysts were not required, making CDCBs cost effective. Nucleophilic aromatic substitution of carbazole anions generated by treatment with NaH and dicyanobenzenes at room temperature (300 K) yielded the CDCBs. All CDCBs were obtained in high yields of >79%, except 4CzPN and 2CzPN, which were obtained in lower yield (38% and 9%, respectively) because of purification problems. CDCBs were fully characterized by NMR and infrared spectroscopy, high-resolution mass spectrometry and elemental analysis (Methods). They had high thermal stability; for example, in thermogravimetric analysis measurements under nitrogen-flow conditions, 4CzIPN began to sublime at around 450 °C before decomposing.

Ultraviolet–visible absorption and photoluminescence spectra of 4CzIPN in toluene are presented in Fig. 2a. 4CzIPN has intense green emission with a maximum at 507 nm and a high photoluminescence quantum yield (Φ) of 94 ± 2%. The Stokes shift of 4CzIPN is very small; generally, the emission produced by intramolecular charge transfer between donor and accepter units shows a large Stokes shift[17]. Under nitrogen-saturated conditions, the delayed component (TADF), which had a lifetime of $\tau \approx 5.1 \pm 0.5$ μs, was more than two orders of magnitude longer than the prompt component, for which $\tau \approx 17.8 \pm 1$ ns (Supplementary Information). To determine whether the T_1 state was involved in luminescence, the transient photoluminescence and photoluminescence quantum yield of 4CzIPN were measured in toluene under an oxygen atmosphere. When oxygen was bubbled through the solution of 4CzIPN in toluene for 10 min, the lifetime of the delayed component became very short ($\tau \approx 91 \pm 3$ ns), the prompt component became $\tau \approx 6.9 \pm 0.5$ ns and Φ decreased to 10%. These results suggest that 4CzIPN is a TADF material because the delayed fluorescence was substantially quenched by oxygen[4].

The geometry of the S_0 state of 4CzIPN in the gas phase was optimized using DFT and the 6-31G(d) basis set[18]. Geometry optimizations of the S_1 and T_1 states were carried out using time-dependent DFT. All *ab initio* calculations were performed using Gaussian 09 software[19] without symmetry constraints. For 4CzIPN, generalized

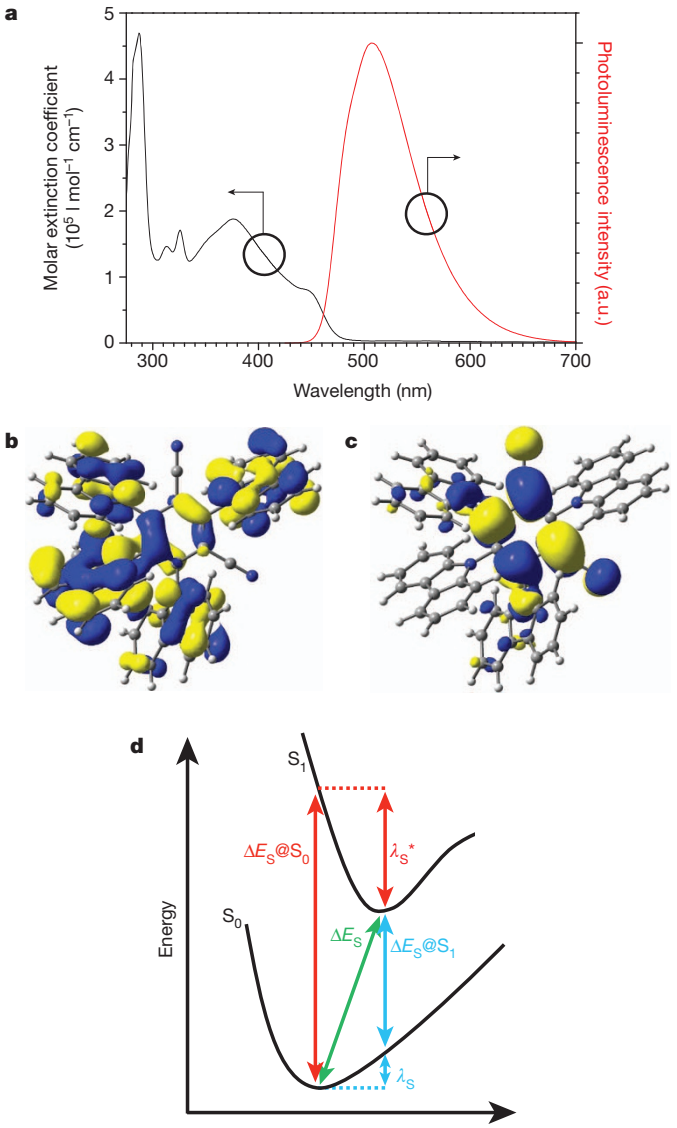


Figure 2 | Photoluminescence characteristics of 4CzIPN. **a**, Ultraviolet–visible absorption and photoluminescence spectra of 4CzIPN in toluene at a concentration of 10^{-5} mol l^{-1}. a.u., arbitrary units. **b**, **c**, Highest occupied NTO (**b**) and lowest unoccupied NTO (**c**) according to the results of time-dependent DFT for the S_1 state of 4CzIPN using the optimized structure of the S_0 state. Colours indicate the different phases of the natural transition orbitals. **d**, Potential energy surfaces, vertical transition energies (ΔE_S@S_0 and ΔE_S@S_1) and relaxation energies (λ_S and λ_S*) for the S_1 state.

gradient approximation functionals underestimated ΔE_{ST} (the experimental value is 83 meV), whereas long-range corrected functionals overestimated it (Supplementary Information). The M06-2X functional[20], a hybrid meta-generalized gradient-approximation functional, yields an intermediate value for ΔE_{ST} and, moreover, by taking into account solvent effects using the polarizable continuum model[21,22], predicts an emission wavelength of 460 nm, which is reasonably close to the experimentally obtained photoluminescence peak at 507 nm in toluene solution. In the following discussion, we use the results calculated for M06-2X/6-31G(d). The highest occupied and lowest unoccupied natural transition orbitals[23] (NTOs) based on the results of time-dependent DFT for the S_1 state of 4CzIPN using the optimized structure of the S_0 state are depicted in Fig. 2b and Fig. 2c, respectively. The highest occupied NTO is delocalized over the four carbazolyl moieties, whereas the lowest unoccupied NTO is centred on the dicyanobenzene moiety, suggesting that the S_1 state has charge transfer character; the carbazolyl groups act as electron donors and the

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dicyanobenzene group acts as an electron acceptor. Steric hindrance between the carbazolyl and dicyanobenzene moieties causes a large dihedral angle of about 60° between the planes of the carbazolyl and dicyanobenzene groups. Consequently, the highest occupied and lowest unoccupied NTOs are spatially well separated, leading to a small ΔE_{ST} and enhanced $T_1 \rightarrow S_1$ reverse ISC.

As stated above, the Stokes shift of the S_1 state of 4CzIPN is small compared with those for typical charge transfer states, suggesting that the $S_1 \leftarrow S_0$ excitation results in limited relaxation of the molecular geometry. We believe that this small Stokes shift is partly related to the presence of the cyano groups. To investigate the effect of the cyano groups on the geometry relaxation of the S_1 and T_1 states of 4CzIPN, we evaluated vertical transition energies (ΔE_S@S_0 and ΔE_S@S_1) and relaxation energies (λ_S and λ_S*) for the S_1 state (Fig. 2d), and compared them with those of the molecule in which the two cyano groups of 4CzIPN are replaced with hydrogen atoms (4CzBz; Supplementary Information). The reorganization energies λ_S for 4CzIPN and 4CzBz were calculated to be 0.27 and 0.83 eV, respectively. This energy is greatly reduced by introducing cyano groups into the electron-accepting unit. Because λ_S represents the degree of geometry relaxation of the S_1 state to the S_0 state, this result suggests that the cyano groups are important in suppressing geometry relaxation in the fluorescent state of 4CzIPN. In addition, λ_S* is also reduced by the presence of cyano groups. Torsional angles of the carbazolyl groups are calculated to be small in the presence of the cyano groups. This limited torsional flexibility can be a major factor in reducing the non-radioactive decay of 4CzIPN. Like that for the S_1 state, the relaxation energy for the T_1 state (λ_T) is markedly reduced by the cyano groups. Thus, it is probable that the cyano groups suppress non-radiative deactivation from the S_1 and T_1 states, leading to the high photoluminescence quantum efficiency of 4CzIPN.

Photoluminescence spectra of the CDCBs in toluene are presented in Fig. 3. The series of CDCBs yielded a wide range of emission colours ranging from sky blue (473 nm) to orange (577 nm). The emission wavelength depends on the electron-donating and -accepting abilities of the peripheral carbazolyl groups and the central dicyanobenzene unit, respectively. Introduction of methyl or phenyl substituents at the 3- and 6- positions of the carbazolyl groups of 4CzTPN induces a shift of the emission maximum to longer wavelengths. Conversely, in the case of 2CzPN, the presence of fewer carbazolyl groups reduces its electron-donating ability and produces a shift of the emission maximum to shorter wavelengths. We measured the photoluminescence quantum yield and transient photoluminescence of CDCBs in toluene under a nitrogen atmosphere, and are summarized in Supplementary Information. For 4CzPN and 4CzTPN Φ is high (74 ± 3% and 72 ± 3%, respectively), whereas for 4CzTPN-Me, 4CzTPN-Ph and 2CzPN it is lower (47 ± 2%, 26 ± 1% and 47 ± 2%, respectively) because of substituent effects or fewer carbazolyl groups. Because the transient photoluminescence of all CDCBs showed both a nanosecond-scale prompt component and a microsecond-scale delayed component, the CDCBs were confirmed to be TADF materials.

Figure 4a shows the photoluminescence decay curves for emission of 4CzIPN at 100, 200 and 300 K in a 5 ± 1 wt% 4CzIPN:4,4′-bis(carbazol-9-yl)biphenyl (CBP) film. The triplet excitons of 4CzIPN are well confined using a CBP host because the T_1 state of CBP is higher in energy than the S_1 state of 4CzIPN. In addition, the fluorescence of CBP is completely quenched by efficient energy transfer between the guest and host molecules. The intense emission observed around $t = 0$ s corresponds to the prompt component, and the long tail is the delayed fluorescence. The prompt component is assigned to the fluorescence of 4CzIPN. The delayed component is attributed to delayed fluorescence occurring via reverse ISC, that is, TADF, because the photoluminescence spectrum of the delayed fluorescence is identical to that of the prompt fluorescence (Fig. 4b).

Figure 4c shows the temperature dependence of Φ for the prompt and delayed components of the film. The two components were resolved by combining the absolute value of Φ estimated using an integrated-sphere photoluminescence measurement system and the temperature dependence of the photoluminescence decay curves (Supplementary Fig. 2). The prompt component increases very slightly as the temperature decreases, indicating the suppression of non-radiative decay from the S_1 state. Conversely, the delayed component decreases monotonically as the temperature decreases because reverse ISC becomes the rate-determining step, similar to the temperature dependence of tin IV fluoride/porphyrin complexes, which are typical TADF emitters[14]. At room temperature (300 K), a high Φ value, of 83 ± 2%, was observed. To evaluate ΔE_{ST} quantitatively, we estimated the activation energy of the reverse ISC rate constant (k_{RISC}) from $\exp(-\Delta E_{ST}/k_BT)$, where k_B is the Boltzmann constant and T is temperature. This rate constant can be estimated from experimentally determined rate constants and the Φ values of the prompt and delayed components at each temperature using[24]

$$k_{RISC} = \frac{k_p k_d}{k_{ISC}} \frac{\Phi_d}{\Phi_p} \quad (1)$$

where k_p and k_d are the rate constants of the prompt and delayed fluorescence components, respectively; k_{ISC} is the ISC rate constant from S_1 to T_1 states; and Φ_p and Φ_d are the photoluminescence quantum yields of the prompt and delayed components, respectively. In Fig. 4d, the values of k_{RISC} calculated from equation (1), assuming that k_{ISC} was independent of temperature, are plotted against $1/T$ for $T = 200$–300 K. From the Arrhenius plot (Fig. 4d), we estimate an activation energy of 83 meV. Therefore, k_{RISC} would be suppressed considerably at low temperatures. However, even at low temperatures, the Φ value of the delayed component is still high, at >40%, implying

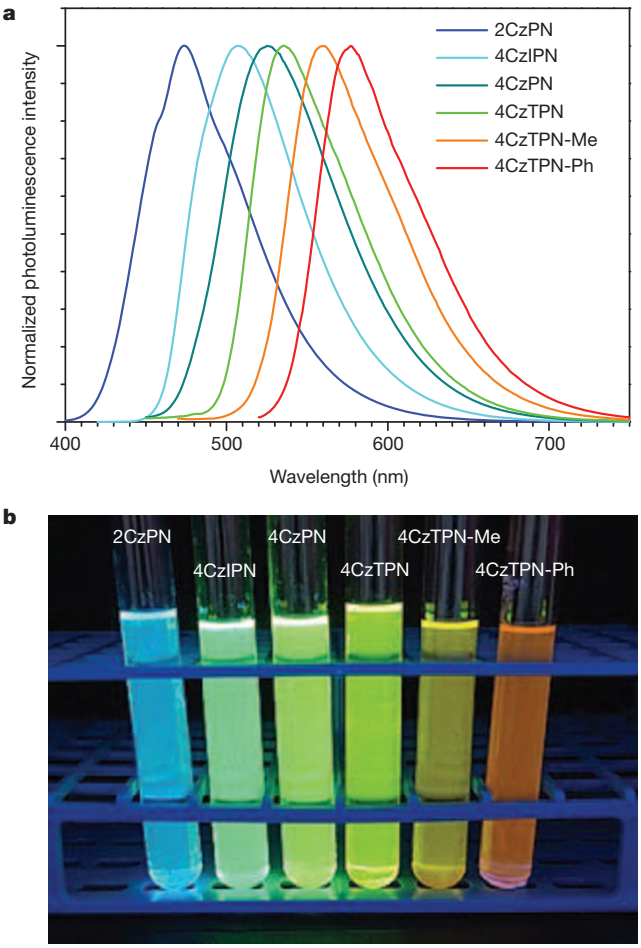


Figure 3 | Photoluminescence of the CDCB series. **a**, Photoluminescence spectra measured in toluene. **b**, Photograph under irradiation at 365 nm.

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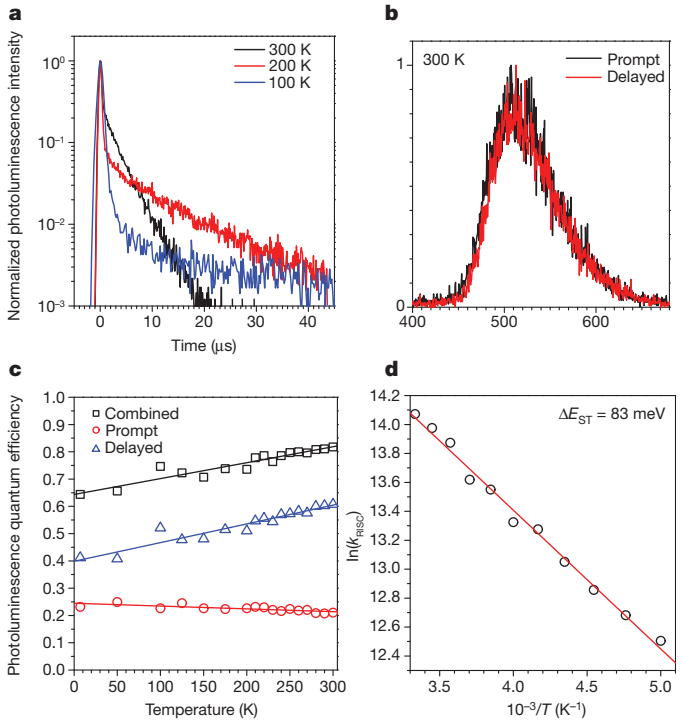


Figure 4 | Temperature dependence of photoluminescence characteristics of a 5 ± 1 wt% 4CzIPN:CBP film. **a**, Photoluminescence decay curves of a 6 wt% 4CzIPN:CBP film at 300 K (black line), 200 K (red line) and 100 K (blue line). The photoluminescence decay curves show integrated 4CzIPN emission. The excitation wavelength of the films was 337 nm. **b**, Photoluminescence spectrum resolved into prompt and delayed components. **c**, Temperature dependence of photoluminescence quantum efficiencies (errors are within 2%) for combined (prompt plus delayed; black squares), prompt (red circles) and delayed (blue triangles) components of 4CzIPN emission for 5 ± 1 wt% 4CzIPN:CBP film. The straight lines are guides for the eye. **d**, Arrhenius plot of the reverse ISC rate from the triplet state to the singlet state of 4CzIPN with k_{ISC} set to $4 \times 10^7\ s^{-1}$. The straight line (least-squares regression) is used to determine the activation energy. The ln(k_{RISC}) errors are within 0.2.

that k_{RISC} is higher than the non-radiative rate constant of the T_1 state. It should be noted that for temperatures less than 200 K the decay curves do not agree with the double-exponential decay model, but are described well by a multi-exponential decay model. This can be explained by the widened ΔE_{ST} distribution caused by inhomogeneous molecular environments at lower temperatures.

We then evaluated the performance of OLEDs containing the CDCB derivatives 4CzIPN (green emission), 4CzTPN-Ph (orange emission) and 2CzPN (sky-blue emission) as emitters. Figure 5 shows the external electroluminescence quantum efficiency of OLEDs containing the CDCB derivatives. To achieve high electroluminescence efficiency in these OLEDs, the T_1 state of the CDCB derivatives must be confined using a host material with a higher triplet energy level. Therefore, we used CBP as a host material in the green and orange OLEDs and used 2,8-bis(diphenylphosphoryl)dibenzo[*b*,*d*]thiophene (PPT) as a host material in the sky-blue OLED. The structures of the OLEDs were composed of multiple layers of indium tin oxide (ITO), 4,4-bis[*N*-(1-naphthyl)-*N*-phenylamino]-biphenyl (α-NPD, 35 nm), 5 ± 1 wt% 4CzIPN or 5 ± 1 wt% 4CzTPN-Ph:CBP (15 nm), TPBi (65 nm), LiF (0.8 nm), Al (70 nm) and ITO (100 nm), α-NPD (40 nm), 1,3-bis(9-carbazolyl)benzene (mCP, 10 nm), 5 ± 1 wt% 2CzPN:PPT (20 nm), PPT (40 nm), LiF and Al. For the green OLED, a very high external electroluminescence quantum efficiency, of 19.3 ± 1.5%, was achieved, which is equivalent to an internal electroluminescence quantum efficiency of 64.3–96.5% assuming a light out-coupling efficiency of 20–30% (refs 25, 26). The orange and sky-blue OLEDs had external electroluminescence quantum efficiencies of

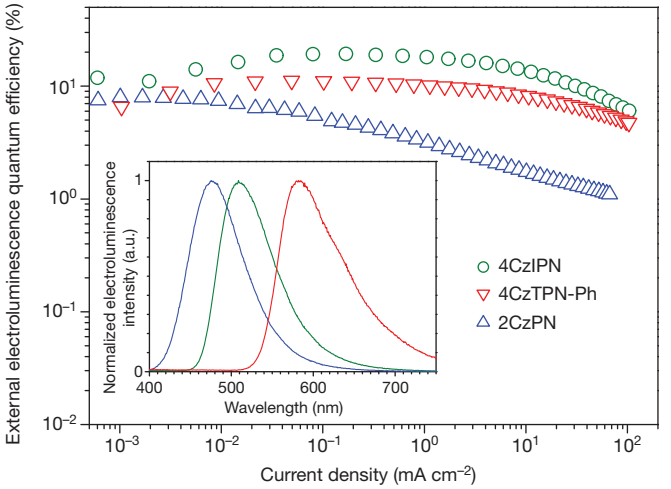


Figure 5 | Performance of OLEDs containing CDCB derivatives. External electroluminescence quantum efficiency as a function of current density for OLEDs containing 4CzIPN (green circles; error within 1.5%), 4CzTPN-Ph (red triangles; error within 1.0%) and 2CzPN (blue triangles; error within 1.0%) as emitters. Inset, electroluminescence spectra of the same OLEDs (coloured accordingly) at a current density of 10 mA cm^{-2}.

11.2 ± 1% and 8.0 ± 1%, respectively, which are also higher than those of conventional fluorescence-based OLEDs.

Finally, we consider the mechanism that drives such efficient reverse ISC without heavy metals. It is generally accepted that the introduction of spin–orbit coupling provided by heavy atoms is required for both ISC and reverse ISC to be efficient. Thus, metal complexes containing heavy metals and aromatic compounds with halogens and carbonyls can promote efficient spin conversion. However, our novel molecular design produces highly efficient spin conversion without needing such atoms. This is because the first-order mixing coefficient between singlet and triplet states (λ) is inversely proportional to ΔE_{ST} (ref. 27):

$$\lambda \propto \frac{H_{SO}}{\Delta E_{ST}} \quad (2)$$

Here H_{SO} is the spin–orbit interaction. It follows from equation (2) that heavy atoms are not required to achieve efficient spin conversion when a molecule possesses a small ΔE_{ST} and H_{SO} is not vanishingly small. This broadens the scope for the molecular design of TADF materials. A detailed strategy for large λ will be clarified on the basis of correlated quantum-chemical calculations[28].

METHODS SUMMARY

Synthesis of carbazolyl dicyanobenzene. The synthesis of CDCB derivatives was performed according to the synthetic method reported previously for pyrrole and fluorobenzene derivatives[29,30]. CDCBs were synthesized by reaction of a carbazolyl anion with a fluorinated dicyanobenzene at room temperature (300 K) for 10 h under a nitrogen atmosphere. CDCBs were purified by column chromatography on silica gel, by reprecipitation or both. CDCBs were further purified by sublimation before photoluminescence and electroluminescence spectra were measured.

Photoluminescence measurements. Organic films for optical measurements were fabricated by thermal evaporation onto clean quartz and silicon substrates under high vacuum ($\sim 7 \times 10^{-4}$ Pa). The photoluminescence spectra of these films were recorded with a spectrofluorometer (FluoroMax-4, Horiba Jobin Yvon), and the photoluminescence quantum efficiencies were measured using an absolute photoluminescence quantum yield measurement system (C9920-02, Hamamatsu Photonics). The transient photoluminescence characteristics were measured (in terms of photon number) under vacuum using a streak camera (C4334, Hamamatsu Photonics). A nitrogen gas laser with a wavelength of 337 nm and a pulse width of approximately 500 ps (MNL200, Lasertechnik) was used as an excitation source. Low-temperature measurements were made using a cryostat (CRT-006-2000, Iwatani Industrial Gases). InGa alloy was applied as an adhesive to ensure good thermal conductivity between the silicon substrate and sample holder.

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OLED fabrication and electroluminescence measurements. The OLED devices were fabricated by thermal evaporation onto clean ITO-coated glass substrates under high vacuum ($\sim 7 \times 10^{-4}$ Pa). Current density/voltage/luminance (J–V–L) characteristics were measured using a semiconductor parameter analyser (E5273A, Agilent) with an optical power meter (1930C, Newport). Electroluminescence spectra of OLEDs were measured (in terms of photon number) using a calibrated multichannel spectrometer (PMA12, Hamamatsu Photonics).

Full Methods and any associated references are available in the online version of the paper.

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Author Contributions H.U. performed the molecular design. H.U. and H.N. performed the synthetic work. K.S. performed the computational experiments. H.U. and K.G. measured the photoluminescence and electroluminescence characteristics of the compounds. All authors wrote the paper. C.A. thought of the TADF concept and supervised the project.

Author Information Reprints and permissions information is available at www.nature.com/reprints. The authors declare no competing financial interests. Readers are welcome to comment on the online version of the paper. Correspondence and requests for materials should be addressed to C.A. (adachi@opera.kyushu-u.ac.jp).

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METHODS

Synthesis of carbazolyl dicyanobenzene. NMR spectra were obtained with a Bruker Biospin Avance-III 500 NMR spectrometer at ambient temperature. Because 4CzTPN, 4CzTPN-Me and 4CzTPN-Ph were not sufficiently soluble in common deuterated solvents, their ^{13}C NMR spectra were not obtained. Infrared spectra were measured with a JASCO FT/IR-620 infrared spectrophotometer. High-resolution mass spectrometry (HRMS) by fast atom bombardment was done using a JEOL JMS-700 spectrometer. Matrix-assisted laser desorption/ionization–time of flight (MALDI–TOF) mass spectrometry was done using a BRUKER Autoflex-III. Elemental analysis was performed with a Yanaco MT-5 elemental analyser. All solvents and chemicals were reagent grade, obtained commercially and used without further purification except as noted. Dry tetrahydrofuran (THF) was purchased from Wako Pure Chemical Industries, Ltd. The synthesis of CDCB derivatives was performed according to the synthetic method reported previously for pyrrole and fluorobenzene derivatives[29,30]. CDCBs were purified by sublimation before photoluminescence and electroluminescence spectra were measured.

We used the following general procedure for the synthesis of tetracarbazolyl dicyanobenzenes. Sixty per cent NaH in oil (0.48 g, 12 mmol) was washed with hexane and then added to a stirred solution of carbazole derivative (10.0 mmol) in dry THF (40 ml) under a nitrogen atmosphere at room temperature. After stirring for 30 min, tetrafluorophthalonitrile, tetrafluoroisophthalonitrile, or tetrafluoroterephthalonitrile (0.400 g, 2.00 mmol) was added. The reaction mixture was stirred at room temperature for 10 h. The reaction was quenched with water (5 ml). The mixture was concentrated under reduced pressure to yield the crude product.

We synthesized 1,2,3,4-tetrakis(carbazol-9-yl)-5,6-dicyanobenzene (4CzPN) according to the general procedure. The crude product was added to water, sonicated for 10 min and then filtered. The residue was purified by column chromatography on silica gel ($CHCl_3$:hexane, 2:1 v/v and then $CHCl_3$:acetone, 2:1 v/v) and then reprecipitated from $CHCl_3$/MeOH to produce 4CzPN as an orange powder (0.602 g, 0.764 mmol, 38%).

The results of our analyses are as follows. Thin-layer chromatography (TLC) ($CHCl_3$:hexane, 4:1 v/v): retardation factor (R_F), 0.20. ^{1}H NMR (500 MHz, DMSO-d_6): δ (p.p.m.) = 7.90 (m, 4H), 7.73 (m, 4H), 7.41 (m, 8H), 7.17–7.12 (m, 8H), 6.76 (m, 4H), 6.62 (m, 4H). ^{13}C NMR (125 MHz, acetone-d_6): δ (p.p.m.) = 141.58, 141.05, 138.75, 137.52, 125.13, 124.03, 123.04, 122.79, 121.06, 120.46, 120.10, 119.29, 119.01, 113.41, 111.17, 111.06. Infrared spectroscopy (KBr): absorption wavelength (cm^{-1}), 2,228, 1,490, 1,457, 1,446, 1,310, 1,223. HRMS (m/z): $[M+H]^+$ calculated for $C_{56}H_{33}N_6$, 789.2766; found, 789.2767; analysis (calculated, found for $C_{56}H_{32}N_6$): C (85.26%, 85.16%), H (4.09%, 4.02%), N (10.65%, 10.55%).

We synthesized 1,2,3,5-tetrakis(carbazol-9-yl)-4,6-dicyanobenzene (4CzIPN) according to the general procedure. The crude product was added to water, sonicated for 10 min and then filtered. The residue was purified by column chromatography on silica gel (hexane:$CHCl_3$, 5:1 v/v and then acetone:$CHCl_3$, 5:1 v/v) and reprecipitated from acetone/$CHCl_3$ to produce 4CzIPN as a yellow powder (1.41 g, 1.79 mmol, 90%).

The results of our analyses are as follows. TLC (acetone:$CHCl_3$, 5:1 v/v): R_F = 0.25. ^{1}H NMR (500 MHz, $CDCl_3$): δ (p.p.m.) = 8.33 (d, J = 7.5 Hz, 2H), 8.06 (d, J = 8.0 Hz, 2H), 7.83 (dd, J = 8.0, 1.5 Hz, 4H), 7.71–7.67 (m, 6H), 7.47 (m, 4H), 7.43 (d, J = 7.5 Hz, 2H), 7.14–7.08 (m, 8H), 6.82 (m, 2H), 6.71 (m, 2H). ^{13}C NMR (125 MHz, acetone-d_6): δ (p.p.m.) = 146.88, 146.20, 141.31, 139.84, 138.82, 137.50, 127.74, 126.41, 125.34, 125.32, 125.03, 124.56, 122.85, 122.32, 121.80, 121.50, 120.98, 120.26, 118.58, 112.99, 111.79, 111.64, 111.36. Infrared spectroscopy (KBr): absorption wavelength (cm^{-1}), 2,233, 1,545, 1,490, 1,479, 1,458, 1,335, 1,309, 1,222. HRMS (m/z): $[M+H]^+$ calculated for $C_{56}H_{33}N_6$, 789.2766; found, 789.2781; analysis (calculated, found for $C_{56}H_{32}N_6$): C (85.26%, 85.22%), H (4.09%, 4.03%), N (10.65%, 10.62%).

We synthesized 1,2,4,5-tetrakis(carbazol-9-yl)-3,6-dicyanobenzene (4CzTPN) according to the general procedure. The crude product was added to water, sonicated for 10 min and then filtered. The residue was purified by column chromatography on silica gel ($CHCl_3$ and then acetone) and reprecipitated from acetone/$CHCl_3$ to produce 4CzTPN as an orange powder (1.50 g, 1.90 mmol, 95%).

The results of our analyses are as follows. TLC (acetone): R_F = 0.05. ^{1}H NMR (500 MHz, DMSO-d_6): δ (p.p.m.) = 7.91 (m, 16H), 7.25 (m, 8H), 7.16 (m, 8H). Infrared spectroscopy (KBr): absorption wavelength (cm^{-1}), 2,235, 1,490, 1,480, 1,455, 1,334, 1,224. HRMS (m/z): $[M+H]^+$ calculated for $C_{56}H_{33}N_6$, 789.2766; found, 789.2781; analysis (calculated, found for $C_{56}H_{32}N_6$): C (85.26%, 85.28%), H (4.09%, 4.11%), N (10.65%, 10.61%).

We synthesized 1,4-dicyano-2,3,5,6-tetrakis(3,6-dimethylcarbazol-9-yl)benzene (4CzTPN-Me) according to the general procedure. The crude product was added to MeOH (200 ml) and then filtered. The residue was washed with water and MeOH to produce 4CzTPN-Me as an orange powder (1.72 g, 1.91 mmol, 95%).

The results of our analyses are as follows. TLC (acetone): R_F = 0.05. ^{1}H NMR (500 MHz, DMSO-d_6): δ (p.p.m.) = 7.81 (d, J = 8.5 Hz, 8H), 7.71 (s, 8H), 7.11 (m, 8H), 2.37 (s, 24H). Infrared spectroscopy (KBr): absorption wavelength (cm^{-1}), 2,241, 1,461, 1,319, 1,281, 1,228. HRMS (m/z): $[M+H]^+$ calculated for $C_{64}H_{49}N_6$, 901.4019; found, 901.4027; analysis (calculated, found for $C_{64}H_{48}N_6$): C (85.30%, 85.39%), H (5.37%, 5.36%), N (9.33%, 9.35%).

We synthesized 1,4-dicyano-2,3,5,6-tetrakis(3,6-diphenylcarbazol-9-yl)benzene (4CzTPN-Ph) according to the general procedure. The crude product was added to water, sonicated for 10 min and then filtered. The residue was purified by column chromatography on silica gel ($CHCl_3$ and then acetone) and reprecipitated from acetone/$CHCl_3$ to produce 4CzTPN-Ph as an orange powder (2.20 g, 1.57 mmol, 79%).

The results of our analyses are as follows. TLC (acetone): R_F = 0.05. ^{1}H NMR (500 MHz, DMSO-d_6): δ (p.p.m.) = 8.37 (d, J = 1.5 Hz, 8H), 8.05 (d, J = 8.5 Hz, 8H), 7.70 (m, 16H), 7.62 (dd, J = 8.5, 1.5 Hz, 8H), 7.45 (m, 16H), 7.36 (m, 8H). Infrared spectroscopy (KBr): absorption wavelength (cm^{-1}), 2,236, 2,228, 1,600, 1,476, 1,456, 1,441, 1,290, 1,226. MALDI–TOF MS (m/z): $[M]^+$ calculated for $C_{104}H_{64}N_6$, 1396.52; found, 1396.66; analysis (calculated, found for $C_{104}H_{64}N_6$): C (89.37%, 89.26%), H (4.62%, 4.53%), N (6.01%, 5.95%).

To synthesize 1,2-bis(carbazol-9-yl)-4,5-dicyanobenzene (2CzPN), 60% NaH in oil (0.96 g, 24 mmol) was washed with hexane and then added to a stirred solution of carbazole derivative (3.34 g, 20.0 mmol) in dry THF (80 ml) under a nitrogen atmosphere at room temperature. After stirring for 30 min, 4,5-difluorophthalonitrile (1.31 g, 8.00 mmol) was added. The reaction mixture was stirred at room temperature for 10 h. The reaction was quenched with water (20 ml). The mixture was concentrated under reduced pressure. The residue was washed with water, purified by column chromatography on silica gel ($CHCl_3$:hexane, 4:1 v/v) and reprecipitated from acetone/hexane to produce 2CzPN as a pale yellow powder (0.310 g, 0.676 mmol, 8.5%).

The results of our analyses are as follows. TLC ($CHCl_3$:hexane, 4:1 v/v): R_F = 0.23. ^{1}H NMR (500 MHz, acetone-d_6): δ (p.p.m.) = 8.73 (s, 2H), 7.79 (m, 4H), 7.38 (m, 4H), 7.13–7.08 (m, 8H). ^{13}C NMR (125 MHz, acetone-d_6): δ (p.p.m.) = 139.86, 139.57, 137.14, 126.76, 124.90, 121.99, 120.96, 116.39, 115.88, 110.74. Infrared spectroscopy (KBr): absorption wavelength (cm^{-1}), 2,236, 1,596, 1,507, 1,478, 1,446, 1,335, 1,221. HRMS (m/z): $[M]^+$ calculated for $C_{32}H_{18}N_4$, 458.1531; found, 458.1529; analysis (calculated, found for $C_{32}H_{18}N_4$): C (83.82%, 83.81%), H (3.96%, 3.80%), N (12.22%, 12.19%).

Photoluminescence measurements. Organic films for optical measurements were fabricated by thermal evaporation onto clean quartz and silicon substrates under high vacuum ($\sim 7 \times 10^{-4}$ Pa). The photoluminescence spectra of these films were recorded with a spectrofluorometer (FluoroMax-4, Horiba Jobin Yvon), and the photoluminescence quantum efficiencies were measured using an absolute photoluminescence quantum yield measurement system (C9920-02, Hamamatsu Photonics). The transient photoluminescence characteristics were measured (in terms of photon number) under vacuum using a streak camera (C4334, Hamamatsu Photonics). A nitrogen gas laser with a wavelength of 337 nm and a pulse width of approximately 500 ps (MNL200, Lasertechnik) was used as an excitation source. Low-temperature measurements were conducted using a cryostat (CRT-006-2000, Iwatani Industrial Gases). InGa alloy was applied as an adhesive to ensure good thermal conductivity between the silicon substrate and sample holder.

Temperature-dependent photoluminescence characteristics of the prompt and delayed components for a 6 wt% 4CzIPN:CBP film were calculated as follows. Supplementary Fig. 2 shows the temperature-dependent lifetime of the prompt component for a 6 wt% 4CzIPN:CBP film. Assuming that the lifetime of singlet excited states is much longer than those of triplet excited states, the respective rate constants of the prompt and delayed fluorescence components (k_p and k_d), in the presence of mutual ISC between the singlet and triplet states, are given by[24]

$$k_p = k_r^S + k_{nr}^S + k_{ISC}$$

$$k_d = k_{nr}^T + \left(1 - \frac{k_{ISC}}{k_r^S + k_{nr}^S + k_{ISC}}\right) k_{RISC}$$

where k_r^S and k_{nr}^S are the radiative and non-radiative rate constants of the singlet state, respectively; k_{ISC} is the ISC rate constant from singlet to triplet states; and k_{nr}^T is the non-radiative rate constant of the triplet state. Therefore, the photoluminescence quantum efficiency of the prompt component is given by

$$\Phi_p = \frac{k_r^S}{k_r^S + k_{nr}^S + k_{ISC}} = k_r^S \tau^S$$

where τ^S is the lifetime of the singlet state. By combining the absolute photoluminescence efficiency estimated using an integrated-sphere photoluminescence measurement system and the temperature dependence of the photoluminescence

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decay curve at $T = 300$ K, we estimated k_r^S to be $1.27 \times 10^7\ s^{-1}$. This allowed us to estimate Φ_p from the temperature-dependent lifetime of the prompt component, assuming that k_r^S is independent of temperature. Using the total photoluminescence quantum efficiency estimated from the temperature dependence of the photoluminescence intensity, the photoluminescence quantum efficiency of the delayed component (Φ_d) can also be estimated. Supplementary Table 4 shows the photoluminescence characteristics of a 6 wt% 4CzIPN:CBP film at each temperature. We estimated k_{RISC} using these experimental data.

OLED fabrication and electroluminescence measurements. The OLED devices were fabricated by thermal evaporation onto clean ITO-coated glass substrates under high vacuum ($\sim 7 \times 10^{-4}$ Pa). Current density/voltage/luminance (J–V–L) characteristics were measured using a semiconductor parameter analyser (E5273A, Agilent) with an optical power meter (1930C, Newport). Electroluminescence spectra of OLEDs were measured (in terms of photon number) using a calibrated multichannel spectrometer (PMA12, Hamamatsu Photonics).

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